Summary of Post-Hoc OOD Detection Methods and Explainability Methods

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1 Introduction

The problem of OOD detection can be formalised as follows. Given a distribution p_{train} from which the data and labels are sampled from i.i.d., there exist many distributions p_{out}, inputs from which may be fed into the model. The goal of the problem is using just p_{train} alongside an auxillary distribution p_{aux}, to train a model f such that $f(x_i) = y_i$ for all $(x_i, y_i) \sim p_{train}$, and $f(x_j) = \text{out}$ for all $x_j, y_j \sim p_{out}$.

2 Methods: OOD Detection

This will focus on post-hoc methods, as opposed to methods that utilise the auxillary dataset to re-train/fine-tune the network to detect OOD samples.

2.1 Maximum Softmax Probability (2016)

$g(x) = \text{argmax}\,\text{softmax}(f(x))$ if $\max\text{softmax}(f(x)) > \tau$ else OOD

Broad Ideas

- Softmax probabilities have poor correspondence to confidence of prediction, because they are very sensitive to noise
- However, they seem to have lower values for out of distribution samples so they provide a reasonable baseline for OOD detection.

Comments:

- Easy to implement
- Vulnerable to adversarial attacks
- Depends on network calibration

2.2 ODIN

Replaces softmax on $f(x)$ with temperature scaled $\frac{f(x)}{T}$. Also backpropogates to find an image which has a higher predicted probability for the maximum probability class. Last step is the same as MSP.

Broad Ideas:

- Temperature scaling can increase the gap in softmax scores between in and out distribution
- Gradient descent on the has a larger effect in increasing the softmax score of the majority class on in-distribution examples than out-distribution examples.
- Find that $||\nabla_x \log S(x;T)||$ is larger in small neighbourhoods of ID images

Comments:

- Effective
- Sensitive to value of T
- Tuning T may be impossible in certain settings and may not generalise to different auxillary sets

2.3 Mahalanobis (2018)

Outperforms ODIN on some tasks. Fits a gaussian distribution to each class conditional distribution. $P(y = c|x) = \frac{\exp((Wf(x)+b)_c)}{\sum_{c'} \exp((Wf(x)+b)_{c'})}$ is the softmax, the class conditional distribution of $f(x)$, $P(f(x)|y = c) = N(\mu_c, \Sigma_c)$. The justification is that Gaussian Discriminant Analysis assuming a single covariance matrix would result in the softmax classifier. The final classifier is created by computing the closest class in the Mahalanobis distance, preprocessing similar to ODIN to increase probability of that class. The confidence score is defined as $\max_c -(f(x) - \hat{\mu}_c)^T \Sigma^{-1} (f(x) - \hat{\mu}_c)$, which is the log pdf of the class-conditional distribution.

Broad Ideas:

- Assuming a class conditional multivariate gaussian with tied covariance, Mahalanobis distance can be used as a confidence score
- Theoretical justification with a heuristic involving gaussian discriminants (reduces to linear discriminant under if covariance is tied)

Comments:

- No sensitivity to hyperparameters
- Restrictive assumptions of shared covariance and gaussian distributed features

2.4 Energy(2020)

$E(f, x) = -T \log(\sum \exp(\frac{f(x)_i}{T}))$. OOD if too small, else ID. Find that temperature scaling is not necessary, degrades performance.

Broad Points:

- Show connection between energy and softmax
- Gradient descent on negative log likelihood decreases the energy associated with the label class and increases the other. Since the energy is dominated by the label class, this is an overall drop
- Softmax creates a biased confidence score which is why it isn't suitable for OOD

2.5 ReAct(2021)

Proposes that clipping activations that are too large should help with OOD detection. Improper normalisation is what causes large spikes, which are a characteristic of OOD inputs and lead to overconfident predictions. Clipping predictions improves separation between ID and OOD distributions using different score functions.

Broad Points:

- Activations at the final layer pre-classification seem to have a heavier tail for the OOD data compared to the ID data.
- Might be intensified by BN, clipping helps control for this
- Do some math to show that the reduction in the embedding and output spaces are both larger for OOD than ID, so clipping increases separation between both types of inputs

2.6 DiCE(2021)

Starts by estimating the average contribution of a neuron to a single class using shapely and uses only the top k weights. Masks away all the unimportant neurons using a top-k approach and uses energy score.

Broad Points:

- Neuron-weight products have heavy tails in both directions, only few important activations
- Sparsifying post-hoc increases the separation over the distribution of energy scores between ID and OOD

2.7 GEM(2022)

Models ID activations as a mixture of k gaussians with different means and the same covariance. Uses the log-sum-exp of the mahalanobis distances from each class-conditional gaussian as score.

Broad Points:

- Mahalanobis distance energy is proportional to the log-likelihood of the density, making it better than raw energy and MSP
- Dimensionality increases OOD uncertainty

2.8 NAP(2022)

Binarises activations from all layers, convolutional and linear and uses minimum hamming distance across classes for OOD detection.

Broad Points:

- Pools convolutional layers into one feature per channel and linear layer neurons into 1 if nonzero else 0
- Creates an atlas for each class and looks at minimum hamming distance from each class
- Refers to a paper which show only few patterns of activation are used by a network on average

2.9 LiNE

Clip the activations like in react. Then, using Shapely, prune to keep only the top k activations and the top k weights in the final layer, then use

Broad Points:

- Only new idea is pruning using shapely instead of magnitude

2.10 ConjNorm(2024)

Estimates $\hat{p}_\theta$ of p_I can be split up as $\sum_{k=1}^{K} \hat{p}_\theta(k|x)\hat{p}_\theta(k)$. Conclusion from the above is that any measure/score function should be $\propto \sum_{k=1}^{K} \hat{p}_\theta(k|x)$, which I'm not sure about because the above itself isn't directly proportional to it. Using the feature extractor of the network to get embedding z, the density can be framed as

$$\hat{p}_\theta(z|k) = \frac{g_\theta(z,k)}{\Phi(k)}$$

g is a density function and Φ is the area under the curve of g, to normalise the probabilities. The paper shows why MSP and Mahalanobis are not proportional to the desired quantity, while GEM, by assuming the distribution is Gaussian is too restrictive in it's assumptions. Then the exponential family of distributions

is defined alongside a notion of bregman divergence using the dual of the legendre that defines the exponential.

$$d_\phi(z, z') = \phi(z) - \phi(z') - \langle z - z', \nabla(\phi(z'))\rangle$$

The bregman divergence is defined using the square of a p-norm, and the final problem left is to solve the partition function estimation. Baselines tried are $\Phi(x) = 1$, estimating $\Phi(x)$ using KDE and using importance sampling, which appears to work best.

Broad Points:

- Most existing metrics are not directly proportional to the class-conditional distribution or have restrictive assumptions
- Estimating the partition function using monte carlo and density using the bregman divergence from the mean

2.11 Greg+ (2024)

Score function ignores gradient and local information, the local area should have ID data if the OOD and ID are indeed different. The idea is to penalise areas which miscategorise ID/OOD as OOD/ID by having energy scores that are too low/high. Often there is a sampling bias in OOD training, because the OOD set is usually larger than the ID set. To rectify this, they cluster the normalised features into n_{ID} clusters, with n_{ID} being the number of in-distribution samples in the minibatch. Then, the lowest energy and highest energy missclassifications are chosen as samples to train the network on. Justification for the whole method is approximate lipschitz continuity.

Broad Points:

- Uses local continuity in the classification i.e. area around a sample should share classification
- Energy score can be used to solve the problem of data imbalance by choosing most valuable samples

2.12 Conformance Constraints

This paper attempts to find constraints to test the reliability of a model on a given tuple. The biggest takeaway from the method is that when taking a PCA of the images, rather than analysing the top k principal components, it's the bottom k principal components which are important because they can be used to define a set of 'invariants' for different datasets.

Broad Points:

- We can do an A/B testing on a trained model by getting the subspaces formed by the bottom k principal components of the ID and OOD dataset and compute a distance between the subspaces.
- Consistent with our work, where a more compact representation is better

3 Explainability/Interpretability Methods

3.1 LIME

Local Explanations for models with only black box access. Approximates a neighbourhood around some datapoint using perturbations and trains a smaller, more interpretable model using these datapoints. The idea is that the importance of different features in the local neighbourhood will be approximated by the smaller model and that can be used to explain predictions.

3.2 SHAP

Tries to estimate a shapely value for each feature, decomposing a prediction into how important each feature is to the prediction amongst all the subsets of features. Needs fairly strong assumptions like feature independence.

3.3 Layer Relevance Propagation

Recursively 'backpropogates' relevance to the input layer using the update

$$R_j^l = \sum_k \frac{a_j w_{jk}}{\sum_{k'} a_j w_{jk'} + \epsilon} R_k^{l+1}$$

Here R_i^l is the relevance of neuron i in layer l, a_j is the value of the jth neuron, w_{jk} is an entry of the weight matrix and ϵ is a smoothing constant Broad points:

- Doesn't work for certain types of layers (local response normalisation, attention etc.)
- Affected by spurious correlations and backdoors

3.4 Saliency Maps

Linear approximation of score/probability of any class using first order taylor expansion and use weight matrix as proxy for pixel-level importance.

- Linearisation would cause massive information loss since the networks are engineered to be very nonlinear
- Pixel level does not seem like a large enough window

3.5 GRAD CAM

At the level of the final convolutional layer, uses the average of the gradients with respect to each feature map as the importance of that feature map to that class and then computes RELU. The visualised combination is a proxy for how important different neurons are to each class for a certain image

3.6 Prototype Relevance Propagation

Builds upon LRP by taking 'prototypes', which are the most 'representative' examples for every class. This is done by relevance propagating to some hidden layer, and computing the similarity of the layer relevance to every element of the dataset. The most similar datapoints in each class become prototypes. Broad Points:

- Not affected by spurious correlations or backdoors
- Directly quantifies similarity of images in embedding space

3.7 KMEx

Ignores last layer, finds a prototypes using the nearest neighbours of the centroids of k-means clusters (performed separately for each class), and uses various methods to assess the quality of the prototypes.

- Finds 'ghosting', where some prototypes are unused almost never happens
- Prototypes are semantically meaningful, because accuracy is high.
- The diversity inside prototypes is large, so the important components of the variance of the embedding space are captured

3.8 Superposition of Neurons and Antrophic's Work on LLMs

The big idea behind superposition is that neural networks are feature extractors with little supervision. Thus, they discover features as necessary but only have a limited number of neurons to represent them. Smaller networks in particular have only a little space, thus in addition to multiple neurons representing the same feature these are squashed into the same neuron. A consequence of this is polysemantic neurons, representing different sparse concepts. Broad points:

- Sparse features (mostly absent in the dataset) are more likely to be represented polysemantically
- Find a geometric property of feature superposition, similar to Thompson problem in chemistry
- Starts by having everything be orthogonal. Then some features are represented by opposite vectors and other geometric structures in increasing order of importance
- Polysemantic neurons may emerge even with efforts to supress superposition (artificially forcing activation sparsity)
- Individual neurons are not always important, multiple neurons make up a feature.

- Dictionary Learning is able to make sense of features in several senses (top-k next predictions upon increasing/supressing the values, change when removing related strings from input tokens etc.)
- Superposition and robustness: If there is no need for superposition and features are disentangled, the representations in the latent spaces for concepts become cleanly low-dimensional without interference from unrelated features. Implications for OOD and proper argument for manifold hypothesis
- Slightly specific to transformers text generation. Haven't understood all the ideas in their entirety, but unlikely that we have to compute to act upon them in any case.

3.9 Work on Explainability in Vision

Broad Points:

- Not individual neurons, classes of neurons that represent 'concepts'
- Optimizing to increase activation of neuron works, but not for combinations of neurons (features)
- Preconditioning by doing gradient descent in the fourier basis seems to help
- Concepts may be better encoded using circuits rather than individual neurons

4 Summary of Methods

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem

Method	**Ideas**	**Comments**
Maximum Softmax Probability	Softmax score as a measure of confidence therefore OOD detection baseline	• States that softmax probability is poorly calibrated, not actual confidence metric • OOD detection, not clear how robustness would work
ODIN	• Temperature scaling with Gradient ascent increases separation between ID and OOD • Gradient of log softmax probability has large norm in the neighbourhood of an ID function	Tuning temperature might not be feasible
Mahalanobis	• Fits class conditional gaussian with single covariance, justifies assumption using LDA • Identical preprocessing to ODIN	• Assumption quite restrictive, our idea of a distribution may be better • Check adversarial examples, distribution seperation can be checked for general robustness

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem (Continued)

Method	**Ideas**	**Comments**
Energy	• Uses energy score with pre-processing to detect OOD, no temperature scaling • Argues why softmax alone is biased and shouldn't be used	Improvement not obvious. Energy scores of our distributions worth checking
ReAct	Activations have strong positive skew on OOD data, so clipping them should increase separation	• Argue that batch norm with improper stats cause abnormal activations, almost the opposite of what we are doing because we are inspecting the tails • Distribution could be used to check if the activations are significantly different across correct and wrong stats
DiCE	Keep neurons with largest magnitude on average, assuming the others are noise	Work out a differentiator
GEM	Sets energy function to mahalanobis distance and uses log energy as score, arguing that it's proportional to log density assuming data density is a mixture of k gaussians with a single covariance matrix	Checking the same metric under our notion of distribution

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem (Continued)

Method	**Ideas**	**Comments**
NAP	Binarises neurons and convolutional layers, creates class atlas and uses minimum hamming distance as score	• Arguably most similar to ours • Doesn't account for noise at all • Reduces each convolutional layer to single filter and purely geometric idea of distance
LiNE	DiCE, replacing largest magnitude with Shapely	• Shapely has been used in CNNs with decent empirical results but skeptical of linear metric • Like ReAct, opposite direction to our work (correcting outliers instead of analysing them)

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem (Continued)

Method	**Ideas**	**Comments**
ConjNorm	• Decomposes true density as sum of class conditionals into class probability • Argues that existing metrics aren't exactly aligned with it or have strong assumptions • Creates metric using bregman divergence of p-norm as density and importance sampling to estimate partition	No ideas for improvement, maybe validation of the density/partition estimator
Conformance Constraints	Least Variant dimensions of a PCA can be used to create invariants for a model and assess trust	Robustness on Limited data, slightly orthogonal approach overall
LIME	Approximates a perturbed neighbourhood around datapoint	Not sure how it relates
SHAP	Feature importance using shapely values	Not sure how it relates
LRP	Interpretability by looking at the most important neurons in the last layer	Not sure how it relates
Saliency Maps	Generalisation of LRP	Not sure how it relates
GRAD-CAM	Generalisation of LRP	Not sure how it relates
PRP	Generalisation of LRP using prototypes, more robust to backdoors and spurious correlation	Not sure how it relates
KMEx	Clustering of embeddings	Our framework suggests the same thing

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem (Continued)

Method	**Ideas**	**Comments**
Anthropic Vision	• Circuits (neurons + weights) are an important unit to study • Neurons may be polysemantic • Monosemantic neurons may lead to polysemantic neurons in the next layer by a phenomenon called superposition	Our work relies on circuits in some sense and talks about their separation at an aggregate level

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem (Continued)

Method	**Ideas**	**Comments**
Anthropic Superposition	• Superposition may be caused by models wanting to represent more directions that dimensions available by exploiting sparsity of features in the data • If features are not sparse, model performs PCA-like collapsing. When they are, it superposes them • Correlated features → orthogonal basis anticorrelated features → opposite ends of same subspace If lack of dimensions → correlated features have similar directions • More superposed models mean more feature interference, lesser adversarial robustness • Privelleged Basis (rotation variant representations) incentivise the features to align with neurons •	

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Table 1: Ideas and Comments from Different Papers on the OOD detection problem (Continued)

Method	**Ideas**	**Comments**
Anthropic LLM	• – Using a sparse autoencoder to extract true high dimensional features – Find some universality in features – Find cleanly interpretable features – 'Features' are more useful for Interpretability than neurons – Features come in clusters –	